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Course Code: ITA0501 -

Computer Vision

Assessment: G2 - ATL - Analytical

Problem Solving.

1) Thresholding and Intensity - level Slicing - PCB Inspection.

Given:

$$r = [25, 60, 95, 130, 165, 200, 235]$$

Intensity band = 90-170

a) Binary Thresholding

Formula:

$$S = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

Output:

$$[0, 0, 0, 255, 255, 255, 255]$$

b) Intensity - level Slicing :-

pixels in the range 90-170 are set to 255;

others remains unchanged

$$[25, 60, 255, 255, 255, 200, 235]$$

c) Number and percentage of highlighted pixels

Highlighted pixels = 3 (95, 130, 165)

$$\text{percentage} = \frac{3}{7} \times 100 = \boxed{42.86\%}$$

d) Mean Intensity:-

Original Mean:

$$\frac{25 + 60 + 95 + 130 + 165 + 200 + 235}{7} = \boxed{130}$$

Sliced output mean:

$$\frac{25 + 60 + 255 + 255 + 255 + 200 + 235}{7} = \boxed{183.57}$$

e) Interpretation:-

Intensity-level slicing is more appropriate because it highlights all pixels in the required color-joint intensity band. Binary thresholding highlights every pixel at a or above 128, including 200 and

235, which are outside the target band.

2) Piecewise linear Contrast stretching - x-ray Image Enhancement.

Given:

$(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$ $L = 256$

a) Equations and Slopes

Segment 1: $0 \leq r \leq 70$

$$S = \frac{20}{70} r = \frac{2}{7} r$$

Slope: $m_1 = 0.286$

Segment 2: $70 < r \leq 140$

$$S = 20 + \frac{230 - 20}{140 - 70} (r - 70)$$

$$\boxed{S = 20 + 3(r - 70). \quad 3r - 190}$$

Slope: $m_2 = 3$

Segment 3: $140 < r \leq 255$

$$S = 130 + \frac{255-230}{255-140}(r-140)$$

Slope:

$$S = 230 + \frac{5}{23}(r-140)$$

$$m_S = 0.2174$$

b) Transformed pixel Values

Input r	Calculation	Output S
30	(217)(30)	8.257
70	Control point	20
90	3(90) - 190	80
110	3(110) - 90	140
140	Control point	230
200	230 + (5/23)(60)	243.04

Rounded output:

[9, 20, 80, 140, 230, 243]

c) Contrast gain factor,

Original diagnostic band width:

$$W_{original} = 140 - 70 = 70$$

Transformed band width:

W1

. transformed = 230 - 20 = 210 ∴

Contrast gain = $\frac{210}{70} = 3$

d) Expansion and Compression

* Segment 1:- Slope 0.286 < 1 - Compresses intensity
* Segment 2:- Slope 3 > 1 - expands intensity
* Segment 3:- Slope 0.2174 < 1 - Compresses intensity.

e) Interpretation

The transformation improves visibility by expanding the diagnostic regions Contrast Functions. However, details Outside the region are Compressed, so that Contrast becomes less visible. No intensity. Clipping occurs for the Specified Expression

3) Spatial filtering - Impulse Noise Removal

Given :-

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

a) 3 x 3 Mean filter :-

$$\text{Mean} = \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$= \frac{689}{9} = \boxed{76.67}$$

b) Weighted - Average filter

Mask :

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Weighted Sum : $22 * (2 \times 222) + (4 \times 255) = 1686$

$$\text{Output} = \frac{1681}{16} = \boxed{105.375}$$

c) Median filter

Sorted Values: 52, 53, 54, 55, 56, 57, 58, 59, 255

The Middle value is: 56

d) Absolute error:- $R = \frac{444}{8} = 55.5$.

Filter	Output	Absolute error
Mean	77.67	22.17
weighted Avg	105.375	49.875
Median	56	0.5

Formula:

Error = |Output - 55.5|

e) Interpretation:-

The Median filter is the most appropriate because it removes the salt-noise impulse and produces the smallest error of 0.5. it also preserves edges better than Averaging filters. Median filtering is particularly effective for salt and pepper noise.

4) Edge detection- Sobel and prewitt Gradient Analysis

Given:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 91 \end{bmatrix}$$

a) Sobel gradient:-

Using the given Sobel masks:

$$G_x = (-10+10)+(-20+180)+(-10+90)$$

$$\boxed{G_x = 240}$$

$$G_y = -40+0+280$$

$$\boxed{G_y = 240}$$

b) Gradient Magnitude and approximation.

Exact Magnitude:

$$M = \sqrt{240^2 + 240^2} = \boxed{339.41}$$

Approximation: $M \simeq |240| + |240| = \boxed{480}$

Percentage error

$$\frac{480 - 339.41}{339.41} \times 100 = \boxed{41.42\%}$$

c) Gradient direction

$$\alpha = \tan^{-1}\left(\frac{240}{240}\right) = \boxed{45^\circ}$$

The gradient direction is 45°, the edge itself is perpendicular to the gradient, at approximately 135°.

d) Prewitt gradient

Using the given Prewitt masks:

$G_x = 0 + 80 + 80 = \boxed{160}$

$G_y = -30 + 0 + 190 = \boxed{160}$

Exact Magnitude:

$$M = \sqrt{160^2 + 160^2} = \boxed{226.27}$$

Approximate:

$$M \approx |160| + |160| = \boxed{320}$$

Percentage error.

$$\frac{320 - 226.27}{226.27} \times 100 = \boxed{41.42\%}$$

c) Edge detection at threshold 250

Operator	Exact Magnitude	Edge detected?
Sobel	339.41	Yes
Prewitt	226.27	No

Interpretation:-

Sobel detected the edge because its Magnitude exceeds 250. Prewitt does not detect it because its Magnitude is below 250. Sobel assigns greater weight to neighbouring pixels, producing a stronger response for this region.

5) Global Thresholding - Iterative and Otsu Segmentation.

Given: total pixels N1 = 32 Intensity levels 0 to 7

a) Global mean intensity,

$$m_G = \frac{\sum r_k n_k}{N}$$

$$m_G = \frac{0(3)+1(6)+2(4)+3(1)+4(1)+5(5)+6(8)+7(4)}{32} = \frac{122}{32} = \boxed{3.8125}$$

b) Iterative global thresholding

Initial threshold:

$$T_0 = m_G = 3.8125$$

Separate pixels into $C_1 : r \leq T$ and $C_2 : r > T$

First iteration:

$$m_1 = \frac{17}{14} = 1.2143$$

$$m_2 = \frac{105}{18} = 5.8333$$

Updated threshold:

$$T_1 = \frac{m_1 + m_2}{2} = \frac{1.2143 + 5.8333}{2} = \boxed{3.5238}$$

Second iteration: The classes remain unchanged.

$$T_2 = \frac{1.2143 + 5.8333}{2} = \boxed{3.5238}$$

$$\therefore \Delta T = |T_2 - T_1| = 0 < 0.5$$

Therefore the final iteration threshold is 3.5238

c) Otsu's between-class Variance

Formula:- $\sigma^2 B = p_1 p_2 (m_1 - m_2)^2$

Threshold k	p_1	p_2	$\sigma^2 B$
2	0.40625	0.59375	5.120
3	0.43750	0.56250	5.251
4	0.46875	0.53125	5.136

d) optimum threshold and binary mapping

The Maximum b/w-class Variance obtained

$$k^* = 3$$

Taking dark objects as in foreground:

Input Intensity	Output	classification
0, 1, 2, 3	1	Object
4, 5, 6, 7	0	Background

Number of object pixels:

$$3+6+4+1=14$$

Object percentage $= \frac{14}{32} \times 100 = 43.75\%$.

c) Comparison and Interpretation.

* Iterative Threshold: T = 3.5238
* Otsu Threshold: K = 3 separating levels 3 and 4
* Both methods produce the same pixel classification for the Histogram
* The Histogram has a reasonable separation between dark and bright intensity groups, So Global Thresholding is suitable for this Bc

Conclusion:-

Otsu's method provides the optimum threshold among the tested values because it gives the highest between-class variance. Global thresholding is effective here, although uneven illumination or overlapping intensity distributions could reduce segmentation accuracy.